

Kingdom of Saudi Arabia

Ministry of Education

Imam Abdulrahman bin Faisal University

College of Science and Humanities-Jubail

Computer science Department

Software Engineering

Term 1 – 2024/2025

Project Proposal Form

• **Title of the Project:**

HealthyCooking: a smart application for discovering recipes and planning meals.

• **Introduction of the Project - Problem Statement:**

Maintaining a healthy diet can be challenging when recipes and meal plans are not tailored to individual needs, time, or preferences. Many existing platforms lack personalization, allergy management, or integrated nutritional tracking making it difficult for users to effectively plan balanced meals. HealthyCooking is an intelligent recipe search and meal planning app designed to address these challenges. It allows users to search recipes by cuisine, meal type, or preparation time, exclude unwanted ingredients, and receive AI-powered suggestions based on what they already have at home. Furthermore, the app offers meal planning and nutrition tools allowing users to track calories and nutrients while creating structured meal plans by combining personalization with health-promoting features. HealthyCooking helps users cook smarter, eat healthier, and achieve their diet goals more effectively.

• **Objectives (Product Characteristics and Requirements):**

1. Make wholesome recipe recommendations based on user preferences and objectives.
2. Take into account essential factors such as calories, available ingredients, and food allergies.
3. Provide a simple and user-friendly interface for easy interaction.
4. Encourage the use of AI tools to create new recipes or alter old ones.
5. Connect to an extensive nutrition and calorie database to provide consumers with precise nutrient data for improved decision-making.
6. To assist customers in following their diet regimens, enable meal scheduling and reminders.

7. Enable users to quickly find recipes based on the amount of time they have available for cooking, making it easier for them to make effective decisions.
8. Provide a platform for users to share their own recipes with the community, fostering interaction and allowing them to discover the creations of others.
9. Allow users to browse a diverse range of recipes from different cuisines around the world, expanding their options and supporting a variety of cooking experiences.

• **Project end user:**

The individuals who are interested in maintaining a healthy lifestyle through balanced nutrient intake. Plus, the general users who want quick meal suggestions based on their preferences.

• **Summary of Project Deliverables:**

HealthyCooking is an intelligent App for finding recipes and meal planning services, designed to help maintain a healthy diet. You can customize it, and maximize its efficiency based on your personal preferences and dietary needs. It provides a built-in nutrition monitoring , such as tracking your daily intake of protein, fiber, etc. Additionally, it offers a search by cuisine, meal type, cooking time, and unwanted ingredients and ingredient-in-hand based AI-powered suggestions. The app recommends healthy recipe ideas based on user choice, calorie consumption, and dietary requirement that are connected to a database for accurate nutrient information. HealthyCooking helps users plan nutritious meals, monitor nutrition, and stick to diet plans. It also supports meal planning, reminders, and AI features for recipe preparation or adjustment. With a simple user-friendly interface, it is perfect for those who are committed to maintaining a healthier lifestyle, allowing individuals to cook smarter, eat better, and achieve their dietary goals with ease.

• **Project Success Criteria:**

The project will be considered successful based on the following:

1. Ease of use and smooth navigation through the application.
2. Stable performance with minimal errors or crashes.
3. User satisfaction with the recipe recommendations.
4. The ability of the AI system to adapt suggestions to user preferences and dietary needs.
5. Reduction in time users spend searching for suitable healthy recipes.

• **Solution Methodology:**

The development of the *HealthyCooking* application will follow a structured and systematic approach to ensure the delivery of a reliable, user-friendly, and efficient solution:

1. Requirements Gathering and Analysis:

Collect and analyze user requirements through surveys and research to identify key features, such as recipe search, AI-powered recommendations, meal planning, and nutrition tracking.

1. UI/UX Design:

Design an intuitive and interactive user interface to provide a smooth user experience, focusing on simplicity, clarity, and responsiveness.

1. Database Development:

Build a well-structured recipe and nutrition database to store accurate information about recipes, ingredients, and nutritional values for better decision-making.

1. AI Model Integration:

Implement and integrate an AI-based recommendation system to provide recipe suggestions tailored to user preferences, dietary needs, and available ingredients.

1. Prototype Development:

Develop the initial version of the mobile application with core functionalities, ensuring proper connectivity between the interface, database, and AI system.

1. Testing and Evaluation:

Conduct functional and usability testing with a sample group of users to evaluate accuracy, performance, and user satisfaction, making necessary adjustments based on feedback.

1. Final Deployment and Refinement:

Deploy the final version of the application after resolving errors, optimizing performance, and enhancing features to ensure a high-quality product.

Date: 27 September 2025

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Approved Not approved Date:

Comments: